

Code-Informed Review of “Evidence Information Gain for LLM Confidence Estimation: A Diffusion Decision Model-Inspired Approach”

Internal review, paper read against the implementation

26 September 2026

This review reads the paper the way an ICLR reviewer would, but every claim was checked against the code and the logs. Items marked `[code: ...]` are facts from the implementation that the paper either omits or contradicts. Part II lists the paper-versus-code discrepancies in one table; Part III gives paste-ready edits.

Part I: Review

Summary

The paper estimates the confidence of a reasoning LLM by treating its chain of thought as a sequence of evidence. The CoT is cut into K equal-length segments; for each segment the paper estimates how much the segment raises the model’s log-probability of its final answer, via either the loss of sampled continuations (white-box) or the agreement rate of 10 sampled continuations (black-box). Levels and first differences of these two signals give a $4K$ -dimensional vector, and a logistic regression maps it to a correctness probability. On Qwen3-8B and DeepSeek-R1-Distill-Qwen-7B, with a leave-one-domain-out protocol, the method reports AUROC ≈ 0.83 – 0.84 against 0.69–0.71 for 10-sample self-consistency and 0.78 for a budget-matched self-consistency.

Strengths

- Simple, clearly motivated idea. The implementation matches the description at the level of the pipeline: one response at $T = 0.2$, 10 continuations per prefix at $T = 1.0$, logistic regression on $4K$ features.
- Cross-domain hold-out is a harder protocol than random splits, and the code really does it: for each of four groups the estimator is trained only on the other benchmarks.
- The gains over SC-10 hold in every held-out group on both models (verified in the per-group tables: +0.10 to +0.17 AUROC).
- The budget-matched self-consistency baseline is the right comparison, and the black-box variant (EIG-SC) tracks the full model within 0.005 AUROC on average.
- Bootstrap standard deviations are computed paired across methods, so the $\pm$ values are comparable between rows.

Weaknesses

1. Cost is never stated, and the budget comparison is not fully matched. CoT-EIG needs one response plus $10(K + 1)$ continuations per question [code: 10 from the question alone, which also serve as SC-10, plus 10 per evidence prefix]: 60 at $K = 5$, 260 at $K = 25$, versus 10 for SC-10 and 25–125 for SC-Budget. No table gives this. SC-Budget exists for Qwen3-8B only; the DeepSeek entry is “—” without explanation [code: no budget logs exist for DeepSeek]. The budget rollouts were generated with a 5,000-token limit while the method’s continuations use 15,000 [code: self_consistency_generation.get_max_new_tokens = 5000 vs. diffusion_decision_model.get_max_new_tokens = 15000], so “computationally matched” is approximate at best; either state it or rerun. In Fig. 4(a) SC-Budget stays at ≈ 0.78 for every K while CoT-EIG-Sum falls to 0.77–0.79 for $K \geq 10$; the text’s “comparable to CoT-EIG-Sum” understates this. At matched rollouts the win is 0.03–0.06 AUROC and comes only from CoT-EIG-Mean.

2. “50 independent runs” is a bootstrap. The estimator is trained once per held-out group; the $\pm$ is the standard deviation over 50 resamples with replacement of the held-out set, evaluated with the fixed estimator [code: BOOTSTRAP_DRAWS = 50, seed 42, in bootstrap_sd]. The main-average $\pm$ is $\sqrt{\sum \text{sd}^2/4}$. Table 1 and §4.3 say “repeated 50 times ...across runs”, which a reviewer will read as training seeds. Nothing in the reported numbers captures training variance or rollout-sampling variance. Bolding 0.84 over 0.83 (DeepSeek, both ± 0.01) marks a difference inside this noise.

3. The reported ECE is not the standard ECE. The paper cites Guo et al. (2017) for ECE, which uses equal-width bins weighted by bin count. The implementation uses 10 equal-count (quantile) bins and an unweighted mean over the non-empty bins [code: calculate_ECE_MCE: np.quantile edges, np.unique, mean over non-empty bins]. For coarse scores such as SC-10 (11 distinct values) several bins collapse. This is a legitimate choice, but it must be stated; a reader recomputing standard ECE will get different numbers. Non-probabilistic baselines are additionally min–max normalized on the test set, which is not a calibration map, so their ECE column is not comparable with the fitted logistic-regression probabilities.

4. Eq. 4 does not describe what the code computes. The paper’s estimator is $-\frac{1}{N_{\text{succ}}} \sum \mathbf{1}[Y^{(n)} = Y] \mathcal{L}(C_i, Y_{>i}^{(n)}) + \mathcal{L}(C_i)$. The code instead takes the mean of the losses of the successful continuations *together with the loss of the prefix itself* appended as one more element, and never subtracts $\mathcal{L}(C_i)$ [code: losses = [x.loss for x in consistency_list if x.accuracy]; losses.append(partial_cot_loss mean(losses))]. Appendix A.2 derives yet a third quantity (the unfiltered $1/N$ average). Three different formulas for one feature; a reviewer who asks for code will find this. Also: when no continuation succeeds, the feature silently becomes the prefix loss alone, and a failed answer extraction counts as a disagreement [code: accuracy = False in the except branch].

5. Sign and definition of Δ . The paper defines $\Delta_{\text{loss}}^i = e_{\text{loss}}^i - e_{\text{loss}}^{i-1}$. The code computes $\text{loss}^{i-1} - \text{loss}^i$ [code: delta_evidence_loss = previous - current] (so that positive means information gain), and sets $\Delta^1 = 0$ [code: channels['delta_loss'].append(0.0)]. The self-consistency feature is Laplace-smoothed, $(m_i + 1)/11$ [code: (true_count + 1)/(len + 1)], and agreement is with the model’s own final answer, not the gold answer. None of this is in the paper. Since the regression is linear, the sign is harmless for results, but the definition should match.

6. Model diversity. Both models are Qwen-derived (DeepSeek-R1-Distill-Qwen-7B is distilled onto Qwen2.5-7B) and both are 7–8B. “Multiple LLMs” is two closely related ones.

7. No trajectory-level baseline. Related work names Mao et al. (2025, 2026), Grünefeld et al. (2026) and Xiong et al. (2026) as the closest work; none is compared to. $P(\text{True})$ and verbalized confidence are absent. Sum-Prob is listed in §4.2 but is not in Table 1 [code: `BASELINE_NAMES` has no Sum-Prob].

8. The feature-contribution analysis mixes two trees and is slightly misdescribed. Figure 2 is a tree with maximum depth 4 and minimum leaf size 10 [code: `DecisionTreeClassifier(max_depth=4, min_samples_leaf=10)`]; Table 2 is a *different*, fully grown tree (depth 20, minimum leaf 1) [code: `feature_importance_nv_5_depth_20_leaf_1.csv` matches Table 2 exactly; the depth-4 tree has only six non-zero features]. The text says “visualized up to depth four”, implying one tree. The claim that SC features “appear in more internal split nodes” is not what Table 2 shows (Gini importance). The label the tree predicts is the regression’s own output ($p > 0.5$) on a random stratified 80/20 split over all eight benchmarks of Qwen3-8B only, not correctness; say so. Most important: the top feature (0.64–0.69) is self-consistency at evidence 1, i.e., agreement after the first $1/K$ of the trace. This is the strongest argument *against* the trajectory framing and the paper does not address it.

9. Countdown and the hold-out description. Countdown is listed among the “eight benchmarks” but appears in no result. In the logs it has only 1–4 incorrect answers per model, so it is train-only [code: `MINORITY_FLOOR = 30` excludes it as a target]. “Each benchmark is used once as the target” is therefore false for Countdown and for MMLU/MMLU-Pro, which are held out jointly. Note also that the sample set changes with K : traces that could not be split into exactly K segments are dropped [code: `if len(log.evidence_list) != number_of_evidence: continue`], so the K ablation is not on a fixed set.

10. Missing experimental details. Not in the paper but fixed in the code: dataset sizes [code: 1,575 / 1,532 questions; test sets of 551/538, 432/431, 185/157, 298/299 per group]; correctness by exact match with `math-verify` for math; 15,000-token generation limit; scikit-learn `LogisticRegression` with defaults (ℓ_2 , $C = 1$), unscaled and unweighted features [code: the reported row is `standardize=False`, `class_weight=None`]; LastRep is the final-layer hidden state averaged over generated tokens [code: `np.mean(last_hidden_state[prompt_length:])`]; SC-10 is the majority share among the 10 continuations from the question alone at $T = 1.0$.

11. “Black-box” is stronger than what is shown. EIG-SC needs the reasoning trace and the ability to resume sampling from a truncated trace at $T = 1.0$ [code: `vLLM` prefill of the prefix]. Commercial reasoning APIs expose neither. “Requires no logits or hidden states” is the accurate claim.

12. Method exposition. Eq. 2 is stated for Y and cites A.1, which proves it for $Y_{>i}$; the step $Y \rightarrow Y_{>i}$ is asserted without argument (it is an upper bound by data processing). “ ≥ 0 ” holds for the mutual information, not for the sampled estimator. “Use the resulting probability as a single estimate of $\mathbb{E}[\log p]$ ” conflates a probability with an expected log-probability. The DDM contributes no fitted quantity; §3.1 is motivation and should be one paragraph.

Presentation

- The main text runs past the page limit (conclusion continues on page 11).
- Conclusion is §5.4 under “Ablation Study”; Appendix A has two untitled subsections; Table 2 sits inside “Proofs”.
- Seven `\cite` where `\citep` is needed; “a LLM” ($\times 2$); “estimation.Six”; “each evidence”/“evidences”; the “Although CoT-EIG already combines...” sentence.
- Fig. 2 is on page 4 but discussed on page 8; Fig. 4’s caption says “baselines” though it includes the proposed method.

Rating

5 as it stands; 6 after Part III; higher with a non-Qwen model. Nothing in the code undermines the main result: the gain over self-consistency is real and reproducible from the logs. What the code exposes is that the paper’s description of the estimator (Eq. 4), the Δ features, the ECE, the “50 runs” and the feature-importance table are each different from what was run. Each is a paragraph to fix; together they are the difference between a reviewer trusting the numbers or not.

Part II: Paper vs. code

Item	Paper says	Code does
Uncertainty ($\pm$)	50 independent runs	50 bootstrap resamples of the held-out set; estimator trained once
Loss feature, Eq. 4	Mean loss of successful continuations, plus $\mathcal{L}(C_i)$	Mean of successful continuation losses with the prefix loss appended as one element; no $+\mathcal{L}(C_i)$ term
Appendix A.2 Δ_{loss} SC feature	Derives unfiltered $1/N$ mean $e^i - e^{i-1}$ “probability”	Neither the paper’s Eq. 4 nor the code $\text{loss}^{i-1} - \text{loss}^i$; $\Delta^1 = 0$ Laplace-smoothed $(m_i + 1)/11$; agreement with the model’s own answer
ECE	Guo et al. (equal-width, count-weighted)	10 equal-count bins, unweighted mean over non-empty bins; min-max on non-probabilities
Table 2 tree	Same tree as Fig. 2 (implied)	Depth 20, min leaf 1; Fig. 2 is depth 4, min leaf 10
Tree label	“confidence labels”	Regression output > 0.5 ; Qwen3-8B only; random 80/20 split
Countdown Hold-out targets	One of eight evaluated benchmarks Each benchmark once	Training only (1–4 errors per model) Four groups; MMLU+MMLU-Pro and GSM8K+MATH500+AIME held out jointly
SC-Budget	Matched compute, both models	Qwen only; 5,000-token cap vs. 15,000 for the method
Sum-Prob Sample set vs. K	Listed baseline Same data for all K (implied)	Does not exist Traces not splittable into K segments dropped per K
LR settings	Not stated	scikit-learn defaults, ℓ_2 , $C = 1$, unscaled, unweighted
LastRep	“final-layer representation”	Mean over generated tokens’ final-layer states
Token limit	Not stated	15,000 new tokens
Correctness	Not stated	Exact match; math-verify for math

Part III: Suggested edits (paste-ready)

Ordered by priority. All numbers are from the logs and code.

1. Table 1 caption (tables/full_auroc_ece.tex)

Where: replace “Results are reported as mean $\pm$ standard deviation over 50 independent runs.”

Results are reported as mean $\pm$ standard deviation over 50 bootstrap resamples of the held-out evaluation set, averaged over the four held-out groups of Section~\ref{sec:cross benchmark generalization}.

2. §4.3, first paragraph

Where: replace “Each benchmark is used once as the target, ...reported across runs.”

This yields four held-out groups: the multiple-choice knowledge domain (MMLU and MMLU-Pro), the mathematics domain (GSM8K, MATH500 and AIME), GPQA, and TruthfulQA. Countdown is used for training only, since both models answer it almost perfectly and it does not contain enough incorrect answers to evaluate a confidence estimator. Reported numbers are the average over the four groups. Standard deviations are computed by resampling each held-out set with replacement 50 times and evaluating the fixed, already-trained estimator on each resample; they therefore reflect evaluation-set uncertainty rather than training variance.

3. §4.1 Evaluation Metrics

Where: replace “Non-probabilistic confidence scores are min-max normalized before computing ECE.”

ECE is computed with 10 equal-count bins over the confidence scores, averaging the absolute gap between mean confidence and accuracy over the non-empty bins. Confidence scores that are not probabilities (losses, entropies and their negations) are min--max normalized to $[0,1]$ on the evaluation set before computing ECE; their ECE should therefore be read as a rescaling rather than a calibrated probability.

4. §3.2, Eq. 4 and its description

Where: replace Eq. 4, Eq. 5 and the sentence “The quantity ...is provided in Appendix A.2.” Pick one of the two options.

Option A: describe what the code computes.

The quantity $\mathbb{E}[\log p(Y_{>i} \mid C_i)]$ is estimated from $N=10$ sampled continuations of the prefix C_i . We keep only continuations whose final answer equals Y , the final answer of the model's original response, and average their sequence losses together with the loss of the prefix itself, which acts as a prior that keeps the estimate defined when no continuation reaches Y :

$$e^i_{\mathrm{loss}} = \frac{1}{N_{\mathrm{succ}} + 1}$$

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\left(
\sum_{n=1}^N
\mathbf{1} \llbracket Y^{(n)}=Y \rrbracket
\mathcal{L}(Y_{>i}^{(n)} \mid C_i) \right)
+
\mathcal{L}(C_i)
\right),
\quad
N_{\mathrm{succ}} = \sum_{n=1}^N \mathbf{1} \llbracket Y^{(n)}=Y \rrbracket,
\label{eq:monte_carlo_estimator}
\end{equation}
where  $\mathcal{L}(Y_{>i}^{(n)} \mid C_i)$  is the loss of the continuation given the
prefix and  $\mathcal{L}(C_i)$  the loss of the prefix. Lower  $e^i_{\mathrm{loss}}$ 
corresponds to higher  $\log p(Y_{>i} \mid C_i)$ .

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Option B: keep Eq. 4 as printed and change the code before the camera-ready. Not feasible in six hours; Option A is recommended. With Option A, delete Appendix A.2 and the reference to it, and delete the “ $\mathcal{L}(C_i, Y_{>i})$ denotes the sequence-level loss of the complete sequence” sentence.

5. §3.3, feature tuple

Where: replace the two Δ definitions in Eq. 6 and add the sentence after it.

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\Delta_{\mathrm{loss}}^i = e_{\mathrm{loss}}^{i-1} - e_{\mathrm{loss}}^i,
\quad
\Delta_{\mathrm{sc}}^i = e_{\mathrm{sc}}^{i-1} - e_{\mathrm{sc}}^i,

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so that both differences are positive when evidence i increases support for Y ; for the first evidence, $\Delta^1_{\mathrm{loss}} = \Delta^1_{\mathrm{sc}} = 0$. The self-consistency estimate is the Laplace-smoothed agreement rate $e^i_{\mathrm{sc}} = (m_i + 1)/(N + 1)$, where m_i is the number of the $N=10$ continuations of C_i whose final answer equals Y .

Where: §3.3, replace “use the resulting probability as a single estimate of $\mathbb{E}[\log p(Y_{>i} \mid C_i)]$.”

use the resulting agreement rate in place of the loss-based estimate as the evidence-level signal.

6. §4.1 Models and Benchmarks

Where: append to the paragraph.

After generation, 575 and 532 questions are retained for Qwen3-8B and DeepSeek-R1-Distill-Qwen-7B, respectively (100--300 per benchmark). Per held-out group, the evaluation sets contain 551/538 (multiple-choice knowledge), 432/431 (mathematics), 185/157 (GPQA) and 298/299 (TruthfulQA) questions for the two models, with the estimator trained on the remaining 1,000--1,400 questions. Responses whose reasoning trace cannot be divided into K segments are excluded for that K .

7. §4.2 Token-level baselines

Where: replace “and token probability (**Mean-Prob**, **Sum-Prob**) are considered.”

and the length-normalized sequence probability (`\textbf{Mean-Prob}`) are considered. Since lower loss and lower entropy indicate higher confidence, these scores are negated so that larger values correspond to higher confidence.

8. §4.2 Self-consistency baselines

Where: replace the whole paragraph.

`\textbf{SC-10}` uses the 10 rollouts sampled from the question alone (i.e., before any evidence is observed) and takes the fraction of rollouts that agree with the majority answer as the confidence. To provide a computationally matched comparison, `\textbf{SC-Budget}` increases the self-consistency sampling budget to \$25\$, \$50\$, \$75\$, \$100\$, and \$125\$ rollouts for $K \in \{5, 10, 15, 20, 25\}$, respectively, where K denotes the number of evidences in the chain-of-thought reasoning trace. These budgets approximately match the number of rollouts our method requires; the budget rollouts were generated with a 5{,}000-token limit. SC-Budget was run for Qwen3-8B only; the corresponding DeepSeek-R1-Distill-Qwen-7B entries are therefore left empty.

9. §4.2, two new paragraphs after “Proposed method”

`\paragraph{Computational cost.}`

For each question, CoT-EIG requires one complete response and $10(K+1)$ sampled continuations: 10 from the question alone (shared with SC-10) and 10 from each of the K evidence prefixes. With $K=5$ this is 60 continuations, compared with 10 for SC-10 and 25 for SC-Budget. Since later prefixes are longer, the continuations are on average shorter than a full response. All continuations are generated once and reused by every variant and feature set, so the ablations in Section~\ref{sec:ablation_features} share the same rollouts.

`\paragraph{Implementation details.}`

Responses and continuations are generated with vLLM with a limit of 15{,}000 new tokens. Final answers are extracted from the response and compared with the reference by exact match, using `\texttt{math-verify}` for the mathematical benchmarks. The confidence estimator is an ℓ_2 -regularized logistic regression (scikit-learn defaults, $C=1$) trained on the unscaled feature vector without class weighting. `\textbf{LastRep}` uses the final-layer hidden states averaged over the generated tokens as its input.

10. §5.2, second paragraph

Where: replace the paragraph.

Figure~\ref{fig:evidence_ablation_models} shows that, on Qwen3-8B, `\textbf{SC-Budget}` reaches an AUROC of about 0.78\$ at every budget, which is above `\textbf{SC-10}` but 0.03\$--0.06\$ below `\textbf{CoT-EIG-Mean}` across all K , and comparable to or slightly above `\textbf{CoT-EIG-Sum}` for $K \geq 10$. Thus, at a matched number of rollouts, the gain of the proposed method comes from CoT-EIG-Mean, and it is moderate but consistent across K .

11. §4.4, second paragraph and Table 2 caption

Where: replace the first paragraph’s “A decision tree is then trained ...is considered.”

A decision tree is then trained on the proposed features to predict these labels, using a random stratified 80/20 split of the combined data of all benchmarks for Qwen3-8B; it thus explains the estimator's decisions, not answer correctness directly. Figure~\ref{fig:feature_importance_tree} shows a tree limited to depth four and a minimum leaf size of 10 for readability. For this analysis, \textbf{CoT-EIG-Mean} with $K=5$ evidences is considered.

Where: replace the second paragraph.

As shown in Figure~\ref{fig:feature_importance_tree}, features from multiple evidences are used as splitting criteria, indicating that confidence estimation relies on information distributed across the reasoning trace. Table~\ref{tab:feature_importance} reports Gini importance for a fully grown tree (maximum depth 20, minimum leaf size 1) fitted on the same data. Self-consistency-based features account for roughly 90% of the total importance, suggesting that they provide the stronger signal. The single most important feature is the self-consistency of the first evidence, i.e., agreement measured after the first $1/K$ of the trace; the remaining evidences add a further $\approx 0.3\%$ of importance, indicating that the trajectory refines an early signal rather than replacing it.

Where: tables/feature_evidence_importance.tex, replace the caption.

\caption{Gini feature importance of a fully grown decision tree (maximum depth 20, minimum leaf size 1) fitted to predict the high/low-confidence label of CoT-EIG-Mean with $K=5$ on the combined data of all benchmarks for Qwen3-8B. Features with zero importance are omitted.}

12. Black-box wording

Where: abstract, replace the last sentence.

Importantly, a variant of our method requires only sampled continuations, without access to token probabilities or hidden states, and achieves confidence estimation performance very close to that of the white-box variant.

Where: contribution bullet 1, replace “and can be applied to LLMs in a black-box setting, without requiring access to their internal representations or parameters,”

and has a variant that requires neither token probabilities nor internal representations, only the ability to sample continuations from a partial reasoning trace,

Where: §3.3, add after “...does not require access to the model’s loss or token-level probabilities.”

It does, however, require sampling continuations from a truncated reasoning trace, which we assume the deployment interface allows.

13. §3.2, $Y \rightarrow Y_{>i}$

Where: replace “we replace the final answer Y with $Y_{>i}$, the continuation of the reasoning chain from e_i through the final answer. This preserves the underlying interpretation while enabling efficient estimation”

we replace the final answer Y with $Y_{>i}$, the continuation of the reasoning chain after e_i up to and including the final answer. Since Y is a deterministic function of $Y_{>i}$, the data-processing inequality gives $I(Y_{>i}; C_i \mid C_{i-1}) \geq I(Y; C_i \mid C_{i-1})$, so the tractable quantity upper-bounds the original one while retaining its interpretation and enabling efficient estimation

14. §3.1, shorten to one paragraph (frees about half a page)

The Diffusion Decision Model (DDM)~\citep{ratcliff1978theory,ratcliff2008diffusion} describes decision making as sequential evidence accumulation under noise, in which a decision is reached when accumulated evidence crosses a boundary. Its central parameter, the \emph{drift rate}, captures the rate and direction of accumulation and reflects the quality of the available evidence~\citep{ratcliff2002estimating,ratcliff2016diffusion}: strong evidence drives fast, consistent accumulation, whereas weak or ambiguous evidence produces slow and erratic accumulation. We borrow this perspective, not the model itself: we treat the segments of a reasoning trace as evidence and quantify each segment's contribution to the final decision, without fitting drift, boundary, or starting-point parameters.

15. Structure and wording

- \subsection{Conclusion} $\rightarrow$ \section{Conclusion}.
- Appendix: first \subsection{} $\rightarrow$ \subsection{Derivation of Eq.~\ref{eq:drift_rate_requation_2}} delete A.2 if Option A of item 4 is taken.
- Move \input{tables/feature_evidence_importance} under a new \section{Feature Importance} after the proofs.
- \cite{ $\rightarrow$ \citep{ in all 7 places; “a LLM” $\rightarrow$ “an LLM” ($\times 2$); “estimation.Six” $\rightarrow$ “estimation. Six”.
- §5.3, replace “Although **CoT-EIG** already combines ...as shown in Figure 5.” with:

\textbf{CoT-EIG}, which combines the loss- and self-consistency (SC)-based features, achieves the strongest overall performance in both AUROC and ECE, while the EIG-SC variant performs very close to it, as shown in Figure~\ref{fig:feature_ablation}.

- Figure 4 caption: “Performance of different confidence estimation baselines” $\rightarrow$ “Performance of the proposed method and the baselines”.
- Abstract and contribution bullet 3: “eight diverse benchmarks and multiple LLMs” $\rightarrow$ “eight benchmarks and two reasoning models”.

16. Limitations sentence

Where: conclusion, insert before the “More broadly” paragraph.

Our evaluation covers two 7--8B reasoning models that share a Qwen lineage and a fixed rollout budget of 10 continuations per evidence; extending the comparison to larger and unrelated model families, and to recent trajectory-level estimators~\citep{mao-etal-2026-confidence,grunefeld2026tracing}, is left for future work.